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Effects of COVID-19 on Rate of Injury and Position-Specific Injury During the 2020 National Football League Season

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Abstract

Costello II, JP, Wagner, JD, Dahl, VA, Cohen, JL, Reuter, AM, and Kaplan, LD. Effects of COVID-19 on rate of injury and position-specific injury during the 2020 National Football League season. *J Strength Cond Res* XX(X): 000–000, 2023—Because of the COVID-19 pandemic, the National Football League (NFL) made changes to its operations for the 2020 season. We hypothesize an increase in the rate of injuries during the 2020 season. Publicly available data were reviewed to identify NFL injuries from the 2015–2020 seasons. Player position, description of injury, date of injury, and injury setting were recorded. $p \leq 0.05$ was considered statistically significant. For the 2020 season, compared with the 2015–2019 seasons, there was an increased risk of injury during the regular season overall relative risk (RR) = 1.308 ($p < 0.05$), week (W)1 RR = 7.33 ($p < 0.05$), W1–6 RR = 1.964 ($p < 0.05$), W7–12 RR = 1.8909 ($p < 0.05$), and during the postseason overall RR = 1.1444 ($p < 0.05$), calculated using analysis of variance. There was an overall increased risk of abdominal or core injuries RR = 1.248 ($p < 0.05$), groin or hip injuries RR = 2.534 ($p < 0.05$), and hamstring injuries RR = 3.644 ($p < 0.05$). There was an increased risk of hamstring injuries in cornerbacks RR = 3.219 ($p < 0.05$) and running backs RR = 1.1394 ($p < 0.05$), hip or groin injuries in guards RR = 1.105 ($p < 0.05$), Achilles tendon injuries in safeties RR = 1.6976 ($p < 0.05$), quadriceps injuries in running backs RR = 1.6191 ($p < 0.05$), and arm injuries in defensive tackles RR = 1.221 ($p < 0.05$). There was an increase in the overall rate of injuries in the 2020 NFL season, both in the regular season and postseason, compared with the 2015–2019 seasons. The overall rate of abdominal or core, groin or hip, and hamstring injuries increased. Specific player positions saw unique increases in rates of injuries. These findings may be due to numerous operational changes implemented, such as reduced in-person training and the elimination of the preseason, leading to suboptimal, sports-specific conditioning and increased risk of musculoskeletal injury.

Key Words: conditioning, medical aspects of sports, epidemiology, statistics, schedule

Introduction

The unprecedented 2020 COVID-19 pandemic had a significant impact on football operations in the National Football League (NFL). Public health and safety concerns during and after the nationwide COVID-19 quarantine led to significant restrictions on team events, including a mandatory work from home order in March of 2020 (5). Typical offseason training including organized team activities (OTAs) and minicamps were canceled and meetings and workouts were transitioned to a virtual platform. Teams were allowed to resume in-person activities at the end of July 2020, later than in-person activities usually occur. Changes also included an altered training camp structure and elimination of the 2020 preseason (6,38).

As a result of COVID-19's effect on players' typical training and preparation for the season, concern for possible increased injury susceptibility due to changes in conditioning has emerged. The inherent safety concerns to players in American football are well appreciated. Approximately 11–81% of athletes experience an injury at some time during their football playing career, and it has been estimated that an injury occurs in up to 97.7% of NFL games (26,33).

Previous research on NFL injury and prevention has shown an increased risk of musculoskeletal injury associated with lack of proper conditioning and increased stress on the body because of rapid increase in workload over the course of days to weeks instead of months (29). Previous studies have also demonstrated players who train and compete at an increased level of fatigue have higher injury rates (4,15,20). Return to play without the typical preseason training, combined with the demands of the NFL season, could contribute to deconditioning injuries.

Recently, limited literature has been published looking at the effects of COVID-19 on NFL injury rates (1–3,30–32,35). Studies performed by Bailey et al., Omari et al., and Sclafani et al. have looked at the effects of COVID-19 on the 2020 regular season, and Baker et al. at the first 4 weeks of the 2020 NFL regular season (2,3,30,35). These studies suggest an increased incidence of injury compared with previous seasons unaffected by COVID-19 (2,3,30,35). However, Platt et al. (32) analyzed the 2020 NFL regular season and found no change in overall incidence of injury

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or incidence of injury by player position compared with previous seasons. Furthermore, Patetta et al. and Allahabadi et al. analyzed the incidence of anterior cruciate ligament (ACL) tears in the 2020 NFL season (1,31). They found an increase in the incidence of ACL tears during the first weeks of the 2020 regular season; however, when these injuries were compared with the number typically experienced during the preseason, there was no increase (1,31).

Because of the varying reports within the literature and lack of extensive analysis, the current study sought to comprehensively evaluate injury rates in the 2020 NFL season. We hypothesize that the 2020 NFL season will have significantly higher rates of injuries compared with previous seasons due to scheduling changes secondary to the COVID-19 pandemic. To the best of our knowledge, the current study is the most exhaustive report on rates of injury during the 2020 NFL season and is the first to examine postseason and offseason injury rates, anatomic injury rates by player position, and change in the proportion of game vs. practice injuries. The purpose of this study was to examine injury rates secondary to schedule changes associated with the COVID-19 pandemic. Findings can be applied to future scheduling and rescheduling of athletic seasons to minimize rates of injury.

Methods

Experimental Approach to the Problem

This study reviewed all injuries to NFL players for the 2015–2020 seasons. We retrospectively reviewed injury transactions, injury reports, physically unable to perform list placements, and injured reserve placements from the NFL offseason, preseason, regular season, and playoffs to identify injuries. All data were publicly sourced through database documentation and media injury reports including NFL.com, Pro-Football-Reference.com, ProsportsTransactions.com, ESPN.com, and other major news sources, found in Appendix 1. As all data were publicly available and de-identified, this study was exempt from institutional review board approval at the University of Miami. An injury was defined as an event that caused a player to miss the next game or caused the player to be listed as do not return in the case of elimination games. All injuries were cross-referenced and required to be reported in a minimum of 2 independent sources for inclusion. Duplicate injuries, sequelae injuries, and injuries from previous seasons were excluded for the purpose of this analysis.

Subjects

All NFL players who suffered injuries while on a roster during the 2015–2020 seasons were included in this study. Data collected included player position, date of injury, description of injury, period in the NFL calendar, and whether the injury occurred in a game or practice setting. All subjects were at least 18 years old.

Procedures

The NFL season was considered to be from March 1st the spring before the season through February 28th after the season. Athlete exposure (AE) was defined as participation by 1 athlete in an individual game or practice in which they were exposed to the possible risk of injury. During both the regular season and playoffs, AEs were calculated assuming each team held 3 live practices and played 1 game each week. Variabilities in the NFL schedule including the number of teams playing each week due to scheduled team bye weeks, COVID-19-related disturbances, games played on days other than Sunday, and variations in allowable roster sizes due to rule changes were considered when calculating AEs for all weeks across all years. Owing to potential player-specific and team-specific variations in offseason training routines, when comparing offseason injuries, the average number of injuries per week was used instead of AEs. For this study, the offseason was considered any injury that happened outside of the period of the preseason, regular season, or postseason, and encompasses events including but not limited to personal training sessions, OTAs, training camps, and minicamps.

Statistical Analyses

All data analysis was performed using SPSS version 28 (SPSS, Chicago, IL). Offseason, regular season, postseason, and overall season injuries were calculated for the 2015–2020 seasons. Preseason injuries were calculated for the 2015–2019 seasons, but not the 2020 season because of the elimination of the 2020 preseason because of the COVID-19 pandemic. Relevant frequency statistics including anatomical location of injury, player position, and rates of game and practice injuries were extracted. Continuous variables were tested for statistical significance using the 2-tailed t test, and categorical variables were compared using the chi-square test. One-way analysis of variance with multiple comparisons was used for comparison of multiple groups, including injury rates across seasons. Relative risk (RR), defined as absolute RR of injury, with 95% confidence interval (CI) was calculated by comparing cohort injury rates during the 2020 season with the 2015–2019 seasons (RR = [cumulative risk of injury incidence in exposure group]/[1,000 athletic exposures] − [cumulative risk of injury incidence in unexposed group]/[1,000 athletic exposures]). Grouped data used for RR calculations passed a normality test and parametric distribution was assumed. $p \leq 0.05$ was considered statistically significant.

Results

Early Season Injuries

A total of 5,597 injuries were identified in the preseason, regular season, postseason, and offseason with 879, 917, 962, 894, 952, and 993 occurring in the 2015, 2016, 2017, 2018, 2019, and 2020 seasons, respectively.

There were 4,256 regular season (W1–W17) injuries identified across all years (Table 1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A410). There was an increased risk of injury overall in the 2020 regular season with RR = 1.308 (95% CI: 0.05911–2.025, $p = 0.0005$), when compared to the 2015–2019 seasons (Table 1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A410). When looking at individual weeks, there was an increased risk of injury occurrence in W1 for the 2020 regular season with RR = 7.33 (95% CI: 4.132–10.528, $p = 0.0024$), compared with the 2015–2019 seasons (Figure 1). There was no increased risk of injury when comparing any other individual weeks for the 2020 season to the 2015–2019 seasons (Figure 1). When the 2020 regular season was broken into thirds, there was an increased risk of injury occurrence in W1–6 with RR = 1.964 (95% CI: 0.1968–3.731, $p = 0.0304$) and W7–12 with RR = 1.8909 (95% CI: 1.1444–2.637, $p = 0.0208$) compared with the 2015–2019 seasons (Figure 2). There was no change in the risk of injury when comparing W13–17 of the 2020 season to the 2015–2019 seasons (Figure 2).

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Table 1

National Football League injuries per 1,000 AE (*n*) during postseason games and practice.*

Year	Wildcard	Divisional	Conference	Super bowl	Total postseason
2015	0.71 (10)	1.13 (11)	0.56 (2)	0.94 (3)	0.84 (26)
2016	0.50 (5)	0.87 (7)	0.50 (2)	0 (0)	0.47 (14)
2017	1.1 (10)	0.80 (5)	0 (0)	0 (0)	0.47 (15)
2018	1.1 (10)	0.81 (6)	0.28 (1)	0.29 (1)	0.64 (18)
2019	1.2 (11)	0.53 (4)	0.79 (3)	0 (0)	0.63 (18)
2015–2019 Mean	0.93 (9.2)	0.83 (6.6)	0.43 (1.6)	0.25 (0.8)	0.61 (18.2)
2020	1.3 (16)	1.1 (9)	1.9 (8)†	0 (0)	1.1 (33)‡

*AE = Athlete exposure; CI = confidence interval; RR = relative risk.
†Statistically significant increase in conference injury when adjusted for athletic exposure (RR = 1.535; 95% CI: 1.6984–3.372; $p = 0.0004$).
‡Statistically significant increase in all total postseason games combined (RR = 1.4871; 95% CI: 1.0923–1.8819; $p = 0.0158$).
*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$, *ns* = nonsignificant.

Postseason Injuries

There were 124 postseason (wild card, divisional, conference, and Super Bowl) injuries identified across all years (Table 1). There was an increased risk of injury occurrence in the 2020 postseason overall with RR = 1.1444 (95% CI: 1.0802–1.2087, $p < 0.0001$), when compared to the 2015–2019 seasons (Table 1). There was an increased risk of injury in the 2020 conference championship round with RR = 1.08246 (95% CI: 0.0374–1.1274, $p = 0.0004$) when compared to the 2015–2019 seasons (Table 1). No change in the risk of injury for the 2020 individual wildcard games, divisional games, or the Super Bowl was observed compared with the 2015–2019 seasons (Table 1).

Anatomic Location

The most frequent injuries reported were injuries of the knee (941), ankle (732), hamstring (582), concussion (571), and shoulder (414; Table 2, Supplemental Digital Content 2, http://links.lww.com/JSCR/A410). When comparing rates for specific anatomic locations, for the 2020 season, there was an increase in abdominal or core injuries with RR = 1.248 (95% CI: 0.5080–1.987, $p = 0.0011$), groin or hip injuries with RR = 2.534 (95% CI, 0.6596–4.409, $p = 0.0019$), and hamstring injuries with RR = 3.644 (95% CI: 0.7428–6.545, $p = 0.0143$), relative to the 2015–2019 seasons (Figure 3). There were no other significant changes in the rate of injury for specific anatomic locations for the 2020 season compared with the 2015–2019 seasons (Figure 3).

Player Position

The most frequently injured positions across all seasons analyzed were linebackers (806), wide receivers (749), cornerbacks (698), safeties (547), and running backs or full backs (514; Table 3, Supplemental Digital Content 3, http://links.lww.com/JSCR/A410). When comparing rates of injury among position in 2020, there was an increased risk of injury in cornerbacks with RR = 3.99 (95% CI: 2.908–5.072, $p < 0.0001$) and guards with RR = 1.1913 (95% CI; 1.04049–1.3422, $p = 0.0136$), relative to the 2015–2019 seasons (Figure 4).

When examining rates of specific anatomic injuries among different player positions, there were significant increases in the 2020 season for hamstring injuries in cornerbacks with RR = 3.219 (95% CI: 1.356–5.083, $p = 0.0055$), hip or groin injuries in guards with RR = 1.105 (95% CI: 0.8255–1.384, $p < 0.0001$), Achilles tendon injuries in safeties with RR = 1.6976 (95% CI: 1.3021–2.093, $p = 0.005$), hamstring injuries in running backs with RR = 1.1394 (95% CI: 1.1394–2.312, $p = 0.0231$), quadriceps injuries in running backs with RR = 1.6191 (95% CI: 1.1640–2.074, $p = 0.0173$), and arm injuries in defensive tackles with RR = 1.221 (95% CI: 0.2594–1.167, $p = 0.0078$), when compared to the 2015–2019 seasons (Figure 5). There were no other significant changes in rates of specific anatomic injuries among different player positions for the 2020 season compared with the 2015–2019 seasons.

Practice vs. Game Injuries

When comparing the proportion of injuries that occurred in practices vs. games, there was no significant differences during the 2020 season compared with the 2015–2019 seasons (Table 4, Supplemental Digital Content 4, http://links.lww.com/JSCR/A410).

Preseason and Offseason Injuries

On examining rates of injuries before the start of the regular season (preseason plus offseason), there was a significantly lower risk of injury before the 2020 regular season compared with 2015–2019 with RR = 0.461 (95% CI: 0.000–1.685, $p = 0.0073$; Figure 1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A409). Although injuries before the regular season overall were decreased in the 2020 compared with 2015–2020, when specifically analyzing the offseason alone, there was no significant difference for 2020 compared with 2015–2019 (Figure 1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A409).

Discussion

A notable finding of this study was that during W1 of the 2020 NFL regular season, injury rates were increased when compared to W1 of previous regular seasons (Figure 1). This is consistent with previous literature looking at rates of injury early in the 2020 NFL season (1,3). Previous literature has shown that gradually increasing training load improves fitness and protects against injury (9), whereas acute onset of load leads to increased rates of injury across a variety of sports (14,21). An overall increase in the total amount of injuries during the 2020 season was also seen (Table 1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A410), contrary to the findings by Platt et al. (32). Their study, however, only included data from the previous 2 NFL seasons (2018–2019), compared with the more comprehensive previous 5 NFL seasons (2015–2019) included in the current study. The current study's larger sample size provides a more accurate

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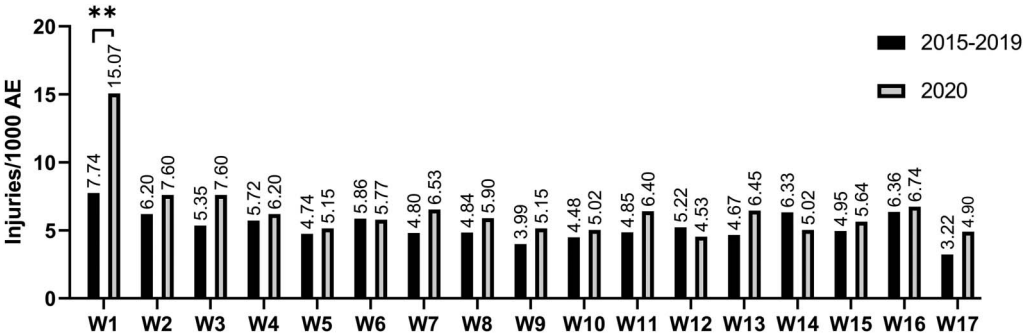

Figure 1. Injury by week for 2015–2019 season compared with 2020 regular season. **Significant increase in injury per 1,000 athletic exposure in the first week of 2020 season (15.07) when compared to the 2015–2019 seasons (7.74), with a relative risk (RR) = 7.33 of injury occurring during the first week (95% confidence interval [CI]: 4.132–10.528; $p = 0.0024$).

representation of the true baseline of injuries in an average NFL season by reducing the possible confounding effects of a single season with an abnormal rate of injuries in the control group.

When the weeks of the regular season were divided into thirds (W1–6, W7–12, and W13–17), W1–W6 and W7–W12 of the 2020 NFL season showed an increased injury rate when compared to those same weeks in the 2015–2019 NFL seasons (Figure 2). Baker et al. (3) reported similar findings in overall injury rates, although this study only examined the first 4 weeks of the 2020 regular season. Patetta et al. and Allahabadi et al. also found an increase in injuries early in the regular season, although these studies were limited to ACL injuries (1,31).

These increases in injuries early in the 2020 regular season may have occurred due to the offseason schedule changes and abrupt start of the NFL season. During the summer, teams were unable to hold official training sessions and could only meet virtually (5). Furthermore, once a return to play protocol was initiated, it included a 20-day acclimatization period with only 14 padded practices and no preseason games (37). These results correlate with findings in other sports and are consistent with the principle that overall training load, regardless of activity, is important to consider when training because increased training load can increase rates of injury (12).

Our study did not see a rise in the rate of overall injuries in the last third of the season (W13–17). These findings are also consistent with previous literature reporting no increase in ACL injuries at the end of the 2020 season (1). The authors speculate that players were properly conditioned by this later point in the season. This supports the theory that the increase in injury rates early in regular season was due to a rapid increase in physical load caused by the absence of typical preseason training.

The current study revealed an overall increase in injury rates during the 2020 NFL postseason when adjusted for AEs (Table 1). Soft tissue injury and muscle breakdown tend to accumulate throughout a season (40), and previous research has demonstrated a correlation between increased play on athlete fatigue and injury (4,15,20). Overuse injuries have also been correlated with the amount of time spent in sport (22). Removal of traditional off-season training and rapid increase in physical load during the 2020 season could be shortening the amount of time necessary to reach a threshold for overuse and fatigue injuries to occur.

When analyzing rates of injuries that occurred before the 2020 NFL regular season (cumulative offseason and preseason injuries), this study found an overall decrease in the rate of injuries per week relative to previous years (Figure 1, Supplemental Digital Content 1, http://links.lww.com/JSCR/A409). These findings suggest that the elimination of preseason games and reliance on training camp style practices to prepare players for the season have a protective effect in the short term due to the decrease in injuries before the season but are harmful in the long term due to the increase in injuries seen overall during the 2020 NFL regular season and 2020 NFL postseason.

Although there was an increased rate of injuries for the 2020 NFL season, this study found no change in the proportion of injuries occurring in practice vs. games compared with previous seasons (Table 4, Supplemental Digital Content 4, http://links.

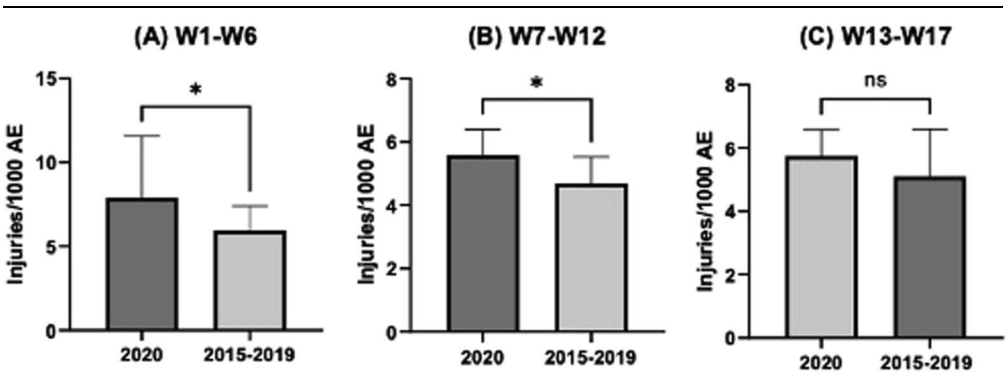

Figure 2: Injury by week for 2015–2019 season compared with 2020 regular season for (A) W1–W6 (relative risk [RR] = 1.964; 95% confidence interval [CI]: 0.1968–3.731; $p = 0.0304$), (B) W7–W12 (RR = 1.8909; 95% CI: 1.1444–2.637; $p = 0.0208$), and (C) W13–W17 ($p = 0.3597$, ns).

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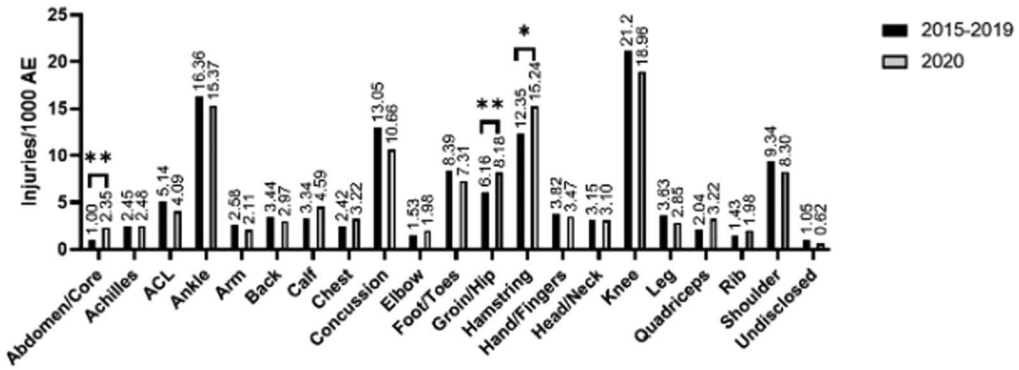

Figure 3. Injury per 1,000 athletic exposure by anatomical location for 2015–2019 and 2020 seasons including preseason, regular season, postseason, and offseason injuries. **Significant increase in abdomen or core injury in 2020 per athletic exposure, relative risk (RR) = 1.248 (95% confidence interval [CI]: 0.5080–1.987; $p = 0.0011$). **Significant increase in groin or hip injury in 2020 per athletic exposure, RR = 2.534 (95% CI: 0.6596–4.409; $p = 0.0019$). *Significant increase in hamstring injury in 2020 per athletic exposure, RR = 3.644 (95% CI: 0.7428–6.545; $p = 0.0143$).

lww.com/JSCR/A410). These findings suggest that although suboptimal conditioning can predispose players to a higher rate of injury, the effects are impartial between the practice and game setting, notable because these settings can vary substantially in their length, composition, and frequency.

We found the most injured anatomical locations among players to be knees, ankles, hamstrings, and shoulders. There was a significant increase among injuries of the abdomen and core, groin and hip, and hamstring injuries across all positions (Table 2, Supplemental Digital Content 2, http://links.lww.com/JSCR/A410). These findings are consistent with limited literature looking at specific soft tissue injuries during the 2020 season (30,35). Although Sclafani et al. found increased rates of soft tissue injuries in the 2020 season, their study used a single previous season (2018–2019) to serve as a baseline. Findings of soft tissue injuries by Omari et al. were also similar; however, their study only analyzed rates of ACL, hamstring, and Achilles injuries. The current study expanded on both studies, using 5 previous NFL seasons as a baseline comparison (2015–2019) and including multiple other soft tissue injuries for comparison. Our study also found no increase in the rate of overall knee injuries or ACL injuries, supporting previous work investigating ACL injuries during the 2020 season (1,31).

The positions most affected by an overall increased rate of injury were cornerbacks and offensive guards (Table 3, Supplemental Digital Content 3, http://links.lww.com/JSCR/A410). Bailey et al. found defensive backs to have had the greatest increase in injury rates (2) which was similar to our findings that cornerbacks, who are a type of defensive back, have increased rates of overall injury in 2020 (Figure 4). Our study did not find an increase in the overall injury rates in safeties, however, who are also defensive backs (Figure 4). No other positions were

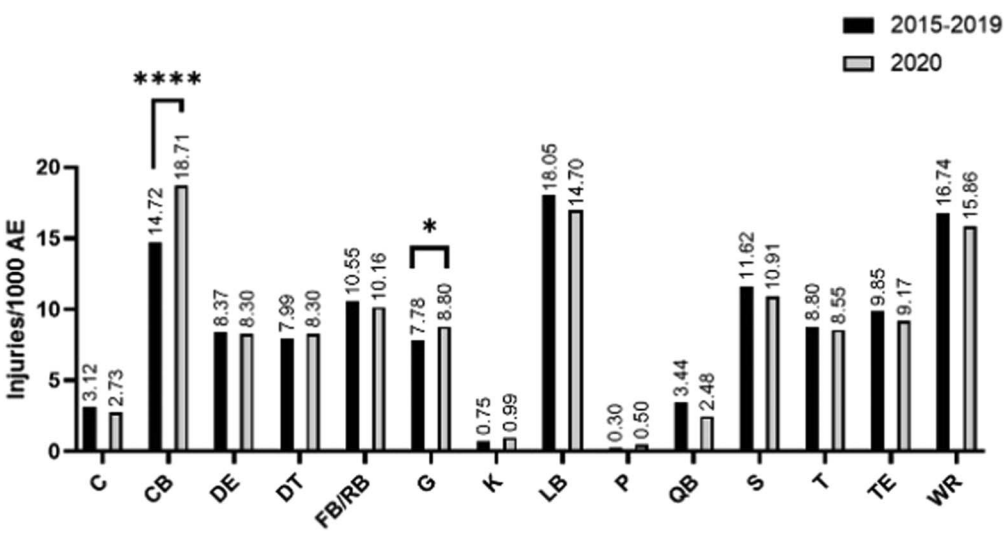

Figure 4. Injury per 1,000 athletic exposure by player position for 2015–2019 and 2020 seasons including preseason, regular season, postseason, and offseason injuries. C = center; CB = cornerback; DE = defensive end; DT = defensive tackle; FB or RB = fullback or running back; G = guard; K = kicker; LB = linebacker; P = punter; QB = quarterback; S = safety; T = tackle; TE = tight end; WR = wide receiver. ****Increased injury per athletic exposure occurring among cornerbacks in 2020 when compared to 2015–2019 (relative risk [RR] = 3.99; 95% confidence interval [CI]: 2.908–5.072; $p < 0.0001$). *Increased injury per athletic exposure occurring among guards in 2020 when compared to 2015–2019 (RR = 1.1913; 95% CI: 1.04049–1.3422; $p = 0.0136$).

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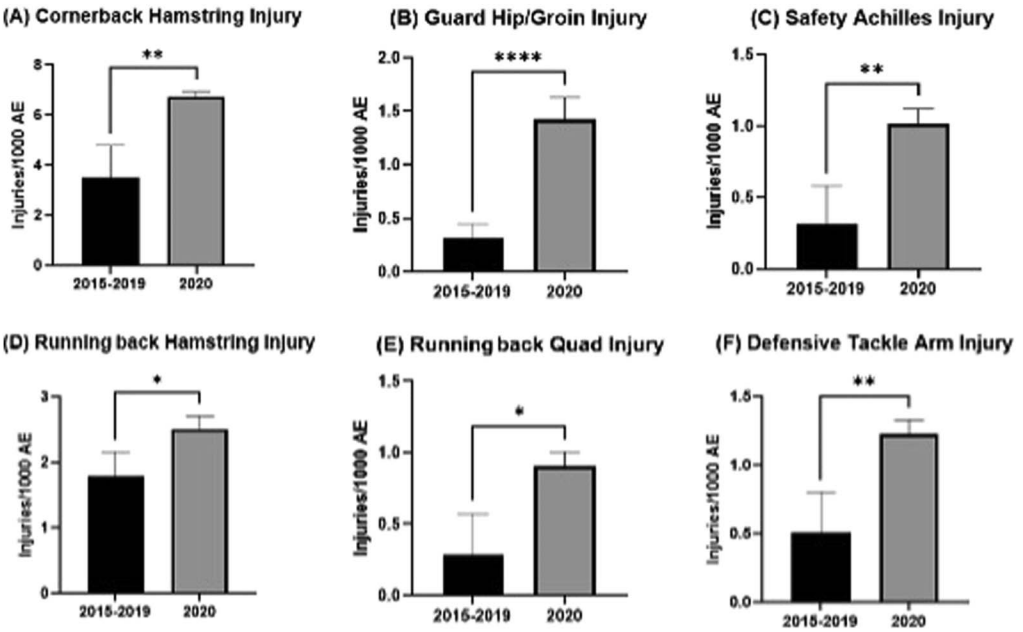

Figure 5. Regular season injury rates per 1,000 athlete exposure (AE) for specific injuries among position groups that demonstrated significant changes during the 2020 season. (A) **Increased rate of hamstring injury in cornerback (CB; relative risk [RR] = 3.219; 95% confidence interval [CI]: 1.356–5.083; $p = 0.0055$). (B) ****Increased rate of hip or groin injury in G (RR = 1.105; 95% CI: 0.8255–1.384; $p < 0.0001$). (C) **Increased rate of Achilles injury in S (RR = 1.6976; 95% CI: 1.3021–2.093; $p = 0.005$). (D) *Increased rate of hamstring injury in RB (RR = 1.1394; 95% CI: 1.1394–2.312; $p = 0.0231$). (E) *Increased rate of quadriceps injury in running back (RB; RR = 1.6191; 95% CI: 1.1640–2.074; $p = 0.0173$). (F) **Increased rate of arm injury in defensive tackle (DT; RR = 1.221; 95% CI: 0.2594–1.167; $p = 0.0078$).

found to have an overall increased rate of injury in 2020 relative to previous seasons.

Interestingly, when analyzing injuries by anatomical location and position together, increases were seen for multiple positions, primarily in soft tissue injuries among skill players (Figure 5). Ward et al. (42) previously found variabilities in metrics such as mechanical load, inertial movement, and overall physical output by player positions in football. Their findings may help explain the positional injury variability in our study because different physical demands may predispose positions to an increased risk for certain injury types, such as soft tissue injuries.

One position and anatomical combination which saw an increase was hamstring injuries among cornerbacks (Figure 5A). Cornerbacks have previously been reported to run greater distances and make more abrupt changes in speed and direction relative to many other positions (7,17,42), leading to an increased baseline risk in hamstring injuries (41). Luxenburg et al. (27) reported defensive backs, including cornerbacks, have among the highest rate of hamstring injuries of all positions. Hamstring injuries may have been particularly affected by the decrease in strength and conditioning time at the beginning of the season because hamstring training has been well documented as a protective factor against injury (19,25,36).

Offensive guards were shown to specifically have an increase in groin or hip injury during the 2020 season (Figure 5B). These players frequently make lateral movements and have been shown to have increased acetabular hip impingement leading to hip pain (13,34). The rapid ramp up in conditioning and elimination of the preseason could predispose these players to increases in these observed hip and groin injuries.

Among safeties, a significant increase in Achilles tears was found (Figure 5C). Similar to cornerbacks, safeties cover great distances on the field with rapid acceleration and deceleration of the position (7), which has been previously identified as a risk factor for Achilles injuries (18). Furthermore, in the 2011 season after the NFL lockout in which there were disruptions to typical offseason training regimens occurred because of disagreement over the collective bargaining agreement (29), increased Achilles ruptures were also seen (29).

During the 2020 season, running backs had increased quadriceps and hamstring injuries (Figure 5D,E). Although one may assume running backs experience rates of injuries proportional to their number of carries, previous research has demonstrated no correlation between the number of carries and rates of injury (23,24). Thus, simply because running backs had fewer games to carry the football during the 2020 season with the elimination of the preseason (5,6), they are not necessarily less likely to be injured. The increase in quadriceps and hamstring injuries observed may be due to differences in position-specific physical demands, with a reduction in training time predisposing them to increases in these injuries (8,10,19,25,36,43).

Rates of arm injuries among defensive tackles were also increased in the 2020 season (Figure 5F). Previous NFL epidemiologic studies have shown a high baseline level of arm injuries among defensive linemen (11,16,39), possibly due to their arms experiencing substantial axial and shearing forces while engaged with offensive linemen (28). Because of the deviations from their typical training and gradual increase in workload throughout the preseason, these players arms could be more vulnerable to injury as a result of these substantial repetitive forces.

The study was retrospective in nature, and as with any study which relies on publicly available data, there was the risk of misreported data. As the authors did not have access to the players electronic medical records, the article is limited in the degree of injury detail reported. In

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addition, although every attempt was made to obtain all relevant information for all players, information, such as specific medical diagnoses or whether the injury occurred in practice or a game, was not available to the public for some players. Substantial efforts were made to account for variabilities in rule changes to roster sizes and game scheduling, but the authors acknowledge there may exist other externalities that we were not able to account for in our analysis.

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Practical Applications

Although the NFL adapted their operations during the COVID-19 pandemic, the changes made to combat the spread of infection may have contributed to the increase in injuries during the 2020 NFL season. Ultimately, alterations to players' traditional offseason training regimens may have led to suboptimal conditioning and preparation for the season. An increased rate of injuries was observed both overall for the regular season, with a notable increase early in the regular season, and overall for the postseason, when compared with previous NFL seasons. Guards and cornerbacks experienced an increase in overall positional injuries, whereas there were also several specific anatomic injuries in unique position groups which increased. These increases in position-specific anatomic injuries included hamstring injuries in cornerbacks and running backs, Achilles injuries in safeties, quadriceps injuries in running backs, hip or groin injuries in guards, and arm injuries in defensive tackles. There was no observed change in the proportion of overall injuries observed in games versus practices. The results of this study highlight the need for proper conditioning and a gradual increase in physical load for athletes across sports. Coaches and athletes alike should be aware of these findings to guide training regimens, practices, and game schedules. Caution should be taken when modifying traditional training schedules, including switching to virtual workouts. These findings are particularly relevant because with the increase in virtual communication and training platforms, individuals and teams will undoubtably attempt to leverage this technology, increasing the time spent away from team facilities during the offseason. This study can additionally be used to understand the epidemiology of NFL injuries by health care professionals. These results enable them to customize treatment protocols that allow for the maximization of playing time and performance for players while reducing risk of injury. Finally, athletic trainers, physical therapists, and coaches can use this information to identify particularly high-risk player positions, as well as their most common anatomic injuries, allowing the creation of optimized workout plans to reduce incidence of specific injuries.

APPENDIX 1

https://www.espn.com/, https://www.cbssports.com/, https://bleacherreport.com/, https://www.foxsports.com/, https://www.nbcsports.com/

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